

DASHBOARD ANALYSIS REPORT

Delivery Performance & Customer Satisfaction

An Analysis of the Olist Brazilian E-Commerce Dataset

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Executive Summary

This report presents the findings of a self-directed Power BI dashboard project built on the Olist Brazilian e-commerce dataset, covering close to 99,000 orders. The dashboard was created to move past a simple revenue-reporting view and instead examine a more specific operational question: how delivery timing affects customer satisfaction, and where that effect is concentrated.

The central insight is this: late deliveries are uncommon, affecting only about 8% of all orders, but their impact on customer perception is severe. Average review scores fall from 4.29 out of 5 for on-time orders to 2.57 out of 5 for late ones, a drop of nearly two full points. This indicates that delivery reliability, while statistically rare as a failure mode, is disproportionately costly to customer satisfaction when it occurs, and therefore deserves targeted operational attention rather than broad, undifferentiated process changes.

Objective & Problem Statement

The dashboard was built to answer a specific operational question rather than to simply display revenue metrics. The guiding questions were:

- What share of orders are delivered later than their estimated delivery date?
- Does late delivery have a measurable effect on customer review scores?
- Which product categories and regions show the highest rates of late delivery?
- Is poor delivery performance concentrated in high-volume markets, or spread evenly across the customer base?

Answering these questions required shifting the dashboard's focus from a standard revenue-and-sales view to one centered on fulfillment reliability and its downstream effect on customer experience.

Data Overview

The analysis uses the publicly available Olist Brazilian E-Commerce dataset, a real-world collection of order records from a Brazilian online marketplace. The dataset spans orders placed between 2016 and 2018 and includes the following core tables and variables:

- Orders: order status, purchase timestamp, approval date, carrier handoff date, actual delivery date, and estimated delivery date.
- Order Items & Products: product category, price, and freight value per item.
- Customers: customer location at state level.

- Payments: payment type and value.
- Order Reviews: customer review score (1 to 5) per order.

Of the roughly 99,000 total orders, about 96,500 have a confirmed delivery date and were used as the base for delivery timing analysis; the remainder were excluded where no delivery was recorded (cancelled, lost, or still in transit at the time the dataset was captured).

Dashboard Design & Structure

The dashboard is organized into three layers, moving from a high-level summary down to specific operational detail:

Top Row — Key Performance Indicators

Five KPI cards give an immediate, at-a-glance read of scale and health: Average Order Value, Total Orders, Total Customers, percentage of orders delivered late, and average review score across all orders. These were deliberately chosen to pair a business-scale metric (orders, customers, order value) with the two metrics central to this analysis (late rate and review score).

Middle Row — Category & Geographic Breakdown

A bar chart shows late delivery rate by product category, filtered to categories with a meaningful order volume to avoid small-sample distortion. Beside it, a scatter plot plots late delivery rate against total order volume by state, allowing high-volume and high-risk markets to be distinguished from low-volume markets with naturally noisier rates.

Bottom Row — Trend and Satisfaction Link

A line chart tracks the monthly trend in late delivery rate, useful for spotting seasonal patterns. Two comparison cards display average review score split by on-time versus late delivery, directly visualizing the central finding. A final bar chart ranks average review score by product category, allowing a cross-check between categories with high late rates and categories with low satisfaction scores.

This structure was chosen deliberately: KPIs answer “what is happening,” the middle row answers “where is it happening,” and the bottom row answers “does it matter, and is it changing over time.”

Key Insights

- **Late delivery is rare but costly to satisfaction.** Only about 8.1% of orders arrive after their estimated delivery date, yet this group shows an average review score of 2.57 out of 5, compared to 4.29 for on-time orders — a gap large enough to materially affect a seller's overall reputation even though it stems from a small fraction of orders.

- **Olist's delivery estimates appear heavily padded.** The large majority of orders arrive well ahead of their estimated delivery date, suggesting the platform builds in significant buffer time when setting customer expectations, rather than running close to its actual logistics capability.
- **Late delivery rates vary meaningfully by product category.** After excluding categories with too few orders to be statistically reliable, categories such as audio, fashion underwear, and technical books show the highest late rates, while categories like baby products and home comfort items perform comparatively better.
- **No simple link between market size and delivery reliability.** Plotting late rate against order volume by state shows no strong pattern of larger markets having systematically better or worse reliability, suggesting causes are more likely tied to specific carriers, categories, or regional logistics infrastructure than to market size alone.
- **Some overlap exists between high late-rate categories and low review-score categories.** Categories such as audio appear among both the highest late-delivery rates and the lower end of average review scores, reinforcing the link between delivery timing and customer perception, though other factors (such as product quality or price disputes) likely also contribute to review scores.

Conclusion

This dashboard demonstrates that customer satisfaction on this platform is not driven primarily by how often deliveries are late, but by how badly satisfaction collapses on the occasions when they are. For a business using this dashboard, the practical implication is that delivery reliability initiatives should be judged by their effect on the tail of the distribution, the relatively small share of late orders, rather than by overall average delivery speed alone. Prioritizing fixes in categories that combine high order volume with above-average late rates, such as electronics and health and beauty, would likely produce the largest improvement in overall customer sentiment per unit of operational effort invested.

Limitations

- The dataset is historical (2016–2018) and does not reflect current operations or any changes Olist may have made since.
- The relationship between late delivery and review score is correlational, not causal. No controlled comparison was performed to isolate delivery timing from other factors that could affect a review, such as product quality or pricing disputes.
- Product categories with very low order counts were excluded from category-level rankings to avoid misleading percentages driven by small sample sizes, meaning some niche categories are not represented in the findings.

- The dashboard reflects a static snapshot of the dataset and does not update in real time.

Future Improvements

- Incorporate seller-level and carrier-level data to identify whether late deliveries are concentrated among specific sellers or shipping partners rather than categories alone.
- Add a forecasting layer to project expected late-delivery rates by category or season, supporting proactive rather than reactive operational decisions.
- Run a more rigorous statistical test (such as a controlled regression) to better isolate the causal contribution of delivery timing to review scores, separate from other satisfaction drivers.
- Layer in geographic distance data (using the dataset's geolocation tables) to test whether shipping distance, rather than state alone, better explains variation in delivery reliability.